

HAO-CHENG KAN

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EDUCATION

Peking University

Master of Computer Applied Technology

Sep 2021 - Jun 2026

- Academic Excellent Awards (Apr 2024)

Lunghwa University of Science and Technology

Master of Business Administration in Information Management

Aug 2015 – Jun 2017

Bachelor of Business Administration in Information Management

Aug 2011 – Jun 2015

- Academic Excellent Awards (Nov 2012, Apr 2013, Oct 2013 and May 2014)
- Minors: Financial Technology; Innovative Entrepreneurship

PROFESSIONAL SKILLS

Typesetting Systems	LaTeX , Microsoft Word
Programming	R, PHP, JavaScript, Java, Python, C Sharp, Visual Basic
Website	HTML5 and CSS
Server	Ubuntu, RedHat, SuSE, Windows Server
Database	MariaDB, Oracle 11g, MS SQL, Hadoop
Project Management	GitHub, Git
Web Framework	Laravel, Angular, React, Vue, Python Flask
DevOps	Docker, AWS
Machine Learning Framework	TensorFlow, PyTorch

RESEARCH/PROJECTS

KoalaMamba: A Federated Method with Vision Mamba UNet for Privacy-Preserving Medical Segmentation (Jul 2026)

- Description: Developed KoalaMamba, a privacy-preserving federated learning framework for medical image segmentation, integrating differential privacy and Vision Mamba UNet backbones to maintain strong segmentation performance while reducing membership inference risks.
- Keywords: Federated Learning, Privacy-Preserving, Medical Image Segmentation

OTTER: Optimized Training with Trustworthy Enhanced Replication via Diffusion and Federated VMUNet for Privacy-Aware Medical Segmentation (Oct 2025)

- Description: Proposed OTTER, a privacy-aware federated medical image segmentation framework using diffusion-based augmentation and Bayesian filtering to improve accuracy while protecting patient data.
- Keywords: Federated Learning, Diffusion Models, Medical Image Segmentation, Privacy-Preserving AI

The Impacts of China's Deleveraging on the Chinese Economy (Jan 2019 – Dec 2019)

- Description: Analyzed China's macroeconomic volatility using DSGE and VAR models, showing that non-financial enterprise deleveraging had short-term negative but limited long-term effects on growth.
- Keywords: Dynamic Stochastic General Equilibrium (DSGE), Vector Autoregression Model (VAR), macroeconomics, deleveraging, business cycle, Basel III, shadow banking system, H-P filter

A3S Academic Achievement Analysis System (Aug 2015 – Sep 2017)

- Description: Developed an interactive student performance analysis system to assess academic, economic, and employment factors and predict post-graduation career development.
- Hao-Cheng Kan & Fang-Ling Lin (10 Dec 2016). The Impact of Student Loans on Academic Performance: A Cluster and Visual Analysis. presented at the 22th CSIM IMP, Taichung, Taiwan.

- Programming languages and software employed: R, RShiny Server, MariaDB, and Ubuntu Server

S-Gate e-Learning and Classroom Management System (Aug 2013 – Aug 2015)

- Description: Developed a mobile e-campus teaching system that streamlined classroom management and student affairs, including Bluetooth-based attendance tracking and real-time mobile interactions such as content delivery, assessments, group discussions, and Q&A.
- Programming languages and software employed: Node.js, JS, PHP, MariaDB, Android, and Ubuntu Server

EXPERIENCE

Hon Hai Precision Industry Co. Ltd

Jun 2024 - Mar 2026

Engineer

Shenzhen

- Embedded Systems and AIOT Technology: Integrate embedded systems with AIOT technology for the development of robots and optimization of industrial production lines.
- Artificial Intelligence and Deep Learning: Utilize artificial intelligence and deep learning techniques to enhance the accuracy and efficiency of industrial inspection, including applications like image segmentation.
- Improvement of Industrial Production Lines: Achieve automation and intelligence in industrial production lines through the use of embedded systems, AIOT, and artificial intelligence technologies, thereby improving production efficiency and product quality.

Peking University

Sep 2021 - Jul 2023

Engineer, Administrative Assistant, and Teaching Assistant

Shenzhen

- Web system maintenance
- Teaching assistant for courses in artificial intelligence and cryptography
- Responsible for performing paperwork in the general office.

OxBridge Consulting Inc

Dec 2018 – Jul 2020

Engineer

Taipei

- Front-end and back-end development, and user interface design
- Conducting systems function tests
- Programming languages and software employed: JavaScript, AngularJS, PHP, CSS, MySql, AWS, and Ubuntu Server

Compulsory Military Service

Dec 2017 – Dec 2018

Corporal

Taichung

- Participating in military trainings and assistance in disaster relief
- Team management, including daily service support, counselling, new recruit orientation, etc.

Lunghwa University of Science and Technology

Aug 2014 – Aug 2016

Lab and Teaching Assistant

New Taipei

- Network equipment implementation and maintenance
- Assistance in the teaching and assignment grading for courses on Linux Server and R programming

JGNet Taiwan

May 2014 – Aug 2014

Network Engineer

Taoyuan

- Broadband network design, installation and maintenance for residential communities

PROFESSIONAL CERTIFICATIONS

- e-Enterprise Data Analyst, 2016
- Red Hat Certified System Administrator (RHCSA), 2015
- Novell Certified Linux Administrator (NCLA 11), 2014
- Oracle Certified Expert (OCE 10g), 2014
- Oracle Certification for Java Programmer (OCJP 6), 2013
- IPMA Certified Project Management Associate (Level D), 2013